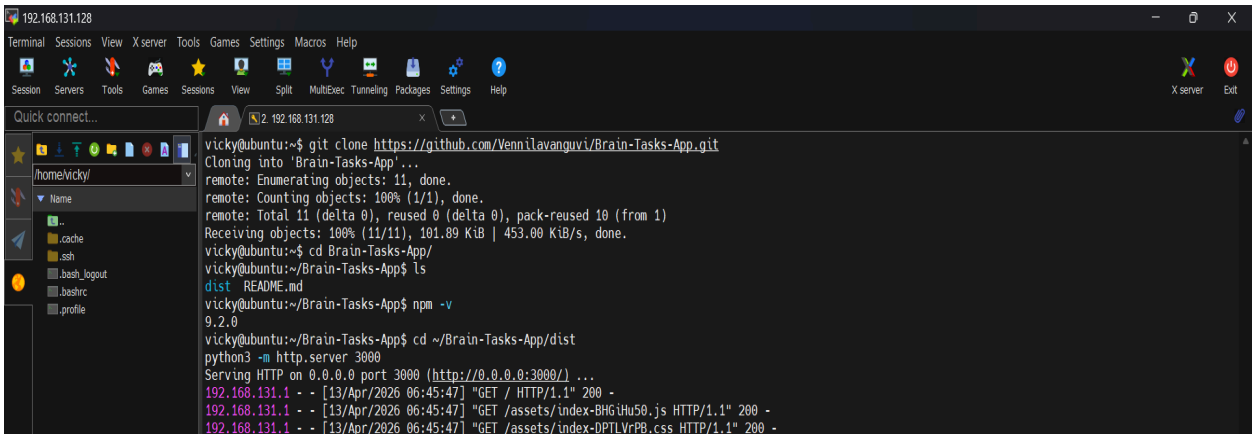
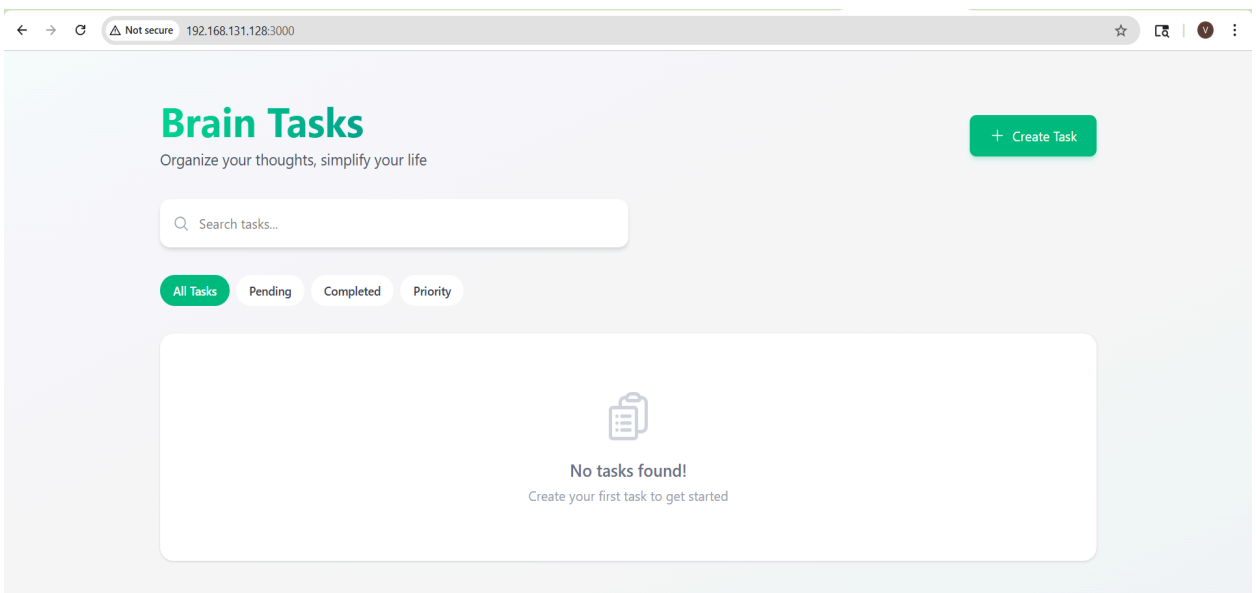
Brain Task Application

Git Repository :

Repository : https://github.com/wikkics/Brain-Tasks-App
Region : us-east-1 (N. Virginia)

Cloned the Brain Tasks App repository and verified the application files locally before containerizing.





Dockerize the Application :

Created a Dockerfile to package the application with Nginx web server into a portable Docker image. Used AWS ECR Public for the Nginx base image to avoid DockerHub rate limits in CI/CD

```
vicky@ubuntu:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu3~24.04.1
vicky@ubuntu:~$ cd ~/Brain-Tasks-App
nano Dockerfile
vicky@ubuntu:~/Brain-Tasks-App$ cat Dockerfile
FROM nginx:alpine

COPY dist/ /usr/share/nginx/html

EXPOSE 80

CMD ["nginx", "-g", "daemon off;"]
vicky@ubuntu:~/Brain-Tasks-App$ docker build -t brain-tasks-app:v1 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon    491kB
Step 1/4 : FROM nginx:alpine
 ---> 582c496ccf79
Step 2/4 : COPY dist/ /usr/share/nginx/html
 ---> 12e4e4187308
Step 3/4 : EXPOSE 80
 ---> Running in 7f698e96f00c
 ---> Removed intermediate container 7f698e96f00c
 ---> 5b25866b8901
Step 4/4 : CMD ["nginx", "-g", "daemon off;"]
 ---> Running in b5a81f37e70f
 ---> Removed intermediate container b5a81f37e70f
 ---> 19bd79d0ec43
Successfully built 19bd79d0ec43
Successfully tagged brain-tasks-app:v1
vicky@ubuntu:~/Brain-Tasks-App$ docker images

IMAGE              ID             DISK USAGE   CONTENT SIZE   EXTRA
brain-tasks-app:v1   19bd79d0ec43       93MB         26.1MB
```

```
vicky@ubuntu:~/Brain-Tasks-App$ docker run -d -p 3000:80 --name brain-app brain-tasks-app:v1
6ca0cf7ad3f8471aa5eb1e47120abdeeee73f92a2c09a0ba921d3242117b7563
vicky@ubuntu:~/Brain-Tasks-App$ docker ps
CONTAINER ID   IMAGE              COMMAND                  CREATED         STATUS         PORTS                                       NAMES
6ca0cf7ad3f8   brain-tasks-app:v1   "/docker-entrypoint.…"   6 seconds ago   Up 6 seconds   0.0.0.0:3000->80/tcp, [::]:3000->80/tcp   brain-app
vicky@ubuntu:~/Brain-Tasks-App$
```

```
vicky@ubuntu:~$ aws --version
aws-cli/2.34.29 Python/3.14.3 Linux/6.8.0-107-generic exe/x86_64.ubuntu.24
vicky@ubuntu:~$ aws configure

Tip: You can deliver temporary credentials to the AWS CLI using your AWS Console session by running the command 'aws login'.

AWS Access Key ID [None]: AKIA4HD2FWIBGY4RUMVK
AWS Secret Access Key [None]: 87UvpjXMHfMkk+ByY+0yjFafI59cuKScqkB2lh0K
Default region name [None]: us-east-1
Default output format [None]: json
```

Container Registry (AWS ECR):

Created an AWS ECR repository and pushed the Docker image to the cloud so Kubernetes can pull it during deployment. AWS ECR is a fully managed private Docker registry — faster, more secure (IAM-controlled), and free of DockerHub rate limits

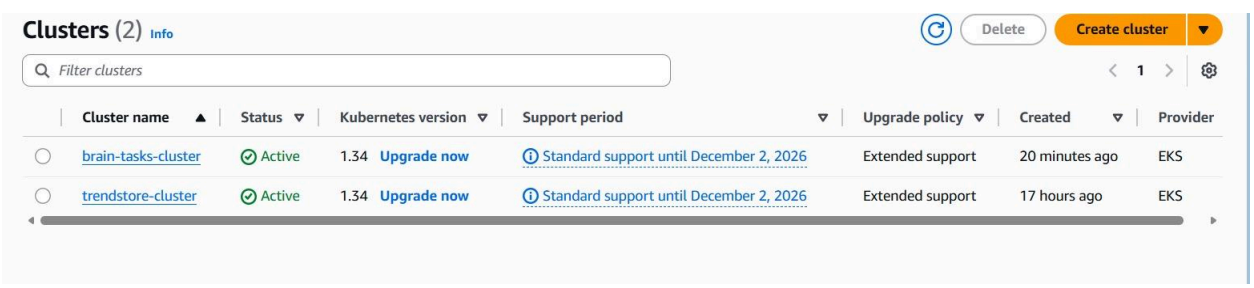
```
vicky@ubuntu:~$ aws ecr create-repository \
  --repository-name brain-tasks-app \
  --region us-east-1
{
    "repository": {
        "repositoryArn": "arn:aws:ecr:us-east-1:839922332162:repository/brain-tasks-app",
        "registryId": "839922332162",
        "repositoryName": "brain-tasks-app",
        "repositoryUri": "839922332162.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app",
        "createdAt": "2026-04-13T06:58:45.573000+00:00",
        "imageTagMutability": "MUTABLE",
        "imageScanningConfiguration": {
            "scanOnPush": false
        },
        "encryptionConfiguration": {
            "encryptionType": "AES256"
        }
    }
}
vicky@ubuntu:~$ aws ecr get-login-password --region us-east-1 | \
  docker login --username AWS \
  --password-stdin \
  123456789012.dkr.ecr.us-east-1.amazonaws.com

WARNING! Your credentials are stored unencrypted in '/home/vicky/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
vicky@ubuntu:~$ docker tag brain-tasks-app:v1 \
  123456789012.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app:v1
```

```
docker tag brain-tasks-app:latest 839922332162.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app:latest
docker push 839922332162.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app:latest
[+] Building 4.2s (7/7) FINISHED                                                                                    docker:default
 => [internal] load build definition from Dockerfile                                                                          0.1s
 => => transferring dockerfile: 157B                                                                                          0.0s
 => [internal] load metadata for public.ecr.aws/nginx/nginx:alpine                                                            3.3s
 => [internal] load .dockerignore                                                                                             0.0s
 => => transferring context: 2B                                                                                               0.0s
 => [internal] load build context                                                                                             0.1s
 => => transferring context: 317.98kB                                                                                         0.0s
 => CACHED [1/2] FROM public.ecr.aws/nginx/nginx:alpine@sha256:72c9b856fee09b02f9e5ee29ab2b55ae9a3c3b59cf094baef1d05f5914d8cb17  0.1s
 => => resolve public.ecr.aws/nginx/nginx:alpine@sha256:72c9b856fee09b02f9e5ee29ab2b55ae9a3c3b59cf094baef1d05f5914d8cb17         0.1s
 => [2/2] COPY dist/ /usr/share/nginx/html                                                                                    0.1s
 => exporting to image                                                                                                        0.4s
 => => exporting layers                                                                                                       0.1s
 => => exporting manifest sha256:d30e953a3d0732a0e86d3428e4059da8b3d891e79135e14fea7413e2a11cd28e                             0.0s
 => => exporting config sha256:f218d0827680b9661b991b8386b0c187fe7495ed41c80a46e13829e17a66a890                               0.0s
 => => exporting attestation manifest sha256:9a9abd04f11137a2ee35aa5497b0e2a4329b7ab0c2f2a570025bc9dfba67b216                 0.0s
 => => exporting manifest list sha256:b025503a8bc68df281b34ca8efd424b12889311c28aa7c2f853cfeb3dc8ade5e                        0.0s
 => => naming to docker.io/library/brain-tasks-app:latest                                                                     0.0s
 => => unpacking to docker.io/library/brain-tasks-app:latest                                                                  0.1s
The push refers to repository [839922332162.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app]
4478572682d3: Pushed
612c0c1df4c5: Pushed
4a8b0b2a5b19: Pushed
4b8840aed9fe: Pushed
aee4e54b3865: Pushed
6a0ac1617861: Pushed
781ff50d2644: Pushed
82736a35d0e7: Pushed
583599bb7d38: Pushed
453da7dbc73e: Pushed
latest: digest: sha256:b025503a8bc68df281b34ca8efd424b12889311c28aa7c2f853cfeb3dc8ade5e size: 856
```

```
vicky@ubuntu:~/Brain-Tasks-App$ kubectl apply -f deployment.yaml
kubectl apply -f service.yaml
kubectl get pods
kubectl get svc
deployment.apps/brain-tasks-deployment unchanged
service/brain-tasks-service unchanged
NAME                                    READY   STATUS    RESTARTS   AGE
brain-tasks-deployment-57c477c5b9-8sp26   0/1     Pending   0          10m
NAME                 TYPE           CLUSTER-IP       EXTERNAL-IP                                                              PORT(S)        AGE
brain-tasks-service  LoadBalancer   10.100.165.205   a50fb9830153340eb8b93606097ebdfa-1501543991.us-east-1.elb.amazonaws.com  80:30370/TCP   10m
kubernetes           ClusterIP      10.100.0.1       <none>                                                                   443/TCP        23m
vicky@ubuntu:~/Brain-Tasks-App$ eksctl scale nodegroup \
  --cluster brain-tasks-cluster \
  --name brain-tasks-nodes \
  --nodes 4 \
  --region us-east-1
2026-04-19 02:48:39 [ℹ]  scaling nodegroup "brain-tasks-nodes" in cluster brain-tasks-cluster
2026-04-19 02:48:44 [ℹ]  initiated scaling of nodegroup
2026-04-19 02:48:44 [ℹ]  to see the status of the scaling run `eksctl get nodegroup --cluster brain-tasks-cluster --region us-east-1 --name brain-tasks-nodes`
vicky@ubuntu:~/Brain-Tasks-App$ kubectl get pods
NAME                                    READY   STATUS    RESTARTS   AGE
brain-tasks-deployment-57c477c5b9-8sp26   0/1     Pending   0          12m
vicky@ubuntu:~/Brain-Tasks-App$ kubectl get pods
NAME                                    READY   STATUS    RESTARTS   AGE
brain-tasks-deployment-57c477c5b9-8sp26   1/1     Running   0          13m
vicky@ubuntu:~/Brain-Tasks-App$
```

Kubernetes on AWS EKS :

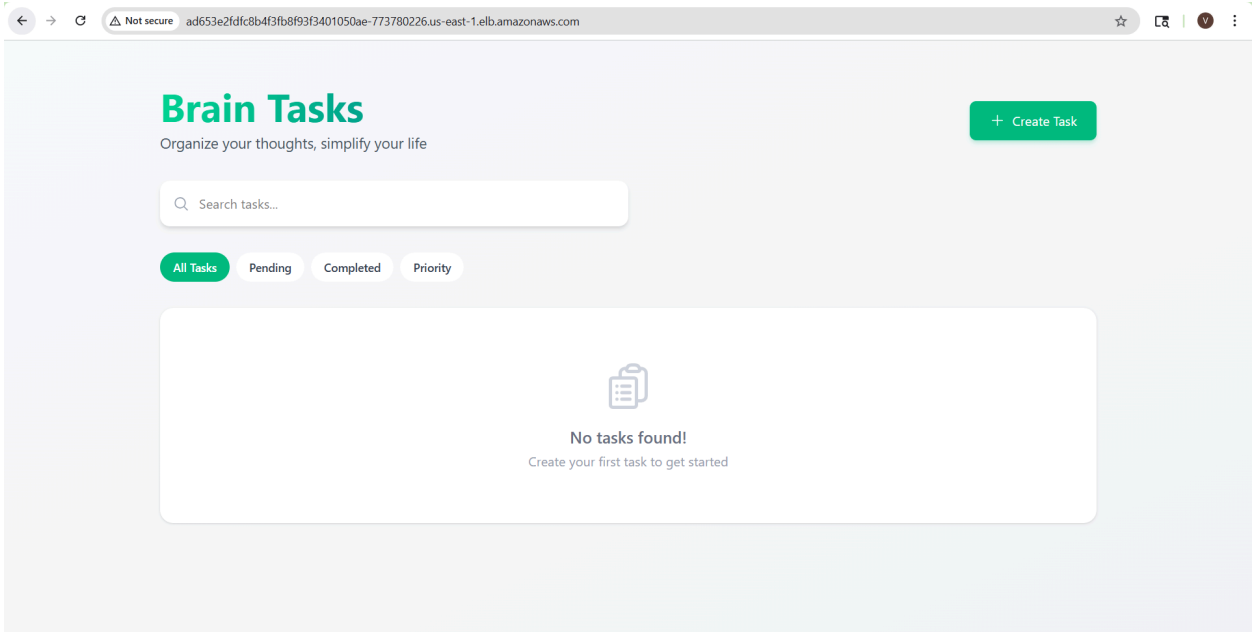
Created a managed Kubernetes cluster on AWS EKS with a t3.small worker node, wrote Kubernetes manifests, and deployed the app with a public LoadBalancer URL.



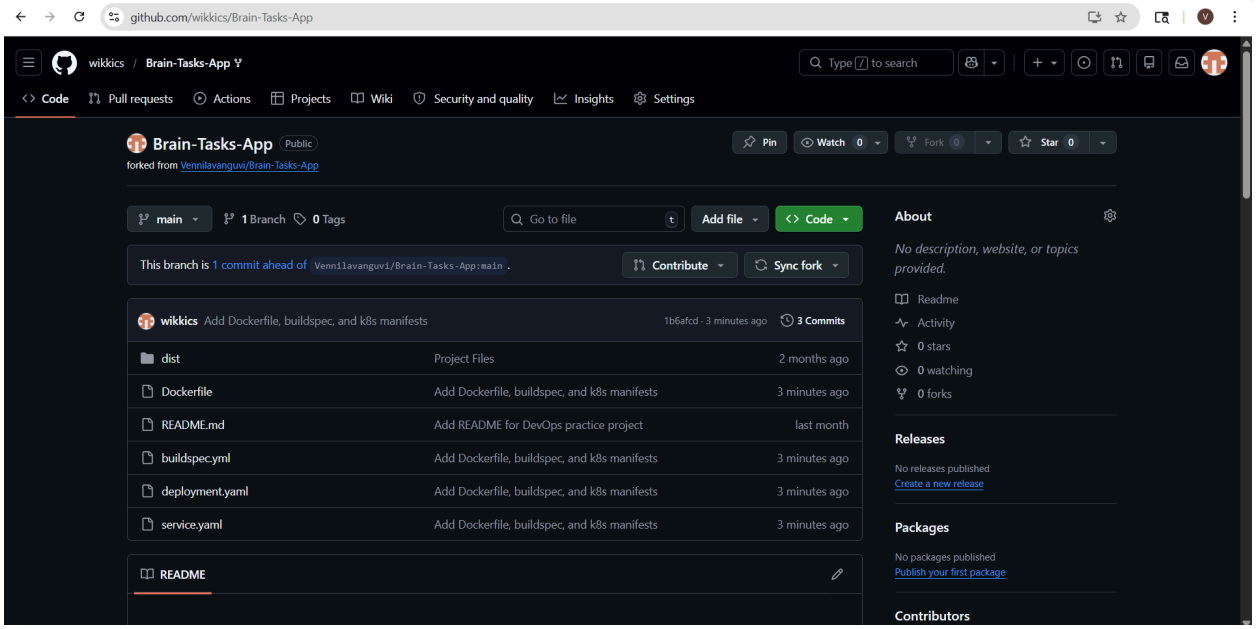
```
2026-04-19 02:22:10 [ℹ]  waiting for CloudFormation stack "eksctl-brain-tasks-cluster-cluster"
2026-04-19 02:23:12 [ℹ]  waiting for CloudFormation stack "eksctl-brain-tasks-cluster-cluster"
2026-04-19 02:24:14 [ℹ]  waiting for CloudFormation stack "eksctl-brain-tasks-cluster-cluster"
2026-04-19 02:25:16 [ℹ]  waiting for CloudFormation stack "eksctl-brain-tasks-cluster-cluster"
2026-04-19 02:26:18 [ℹ]  waiting for CloudFormation stack "eksctl-brain-tasks-cluster-cluster"
2026-04-19 02:26:28 [!]  recommended policies were found for "vpc-cni" addon, but since OIDC is disabled on the cluster, eksctl cannot configure the requested
 permissions; the recommended way to provide IAM permissions for "vpc-cni" addon is via pod identity associations; after addon creation is completed, add all
recommended policies to the config file, under `addon.PodIdentityAssociations`, and run `eksctl update addon`
2026-04-19 02:26:28 [ℹ]  creating addon: vpc-cni
2026-04-19 02:26:29 [ℹ]  successfully created addon: vpc-cni
2026-04-19 02:26:30 [ℹ]  creating addon: kube-proxy
2026-04-19 02:26:31 [ℹ]  successfully created addon: kube-proxy
2026-04-19 02:26:31 [ℹ]  creating addon: coredns
2026-04-19 02:26:32 [ℹ]  successfully created addon: coredns
2026-04-19 02:28:40 [ℹ]  building managed nodegroup stack "eksctl-brain-tasks-cluster-nodegroup-brain-tasks-nodes"
2026-04-19 02:28:43 [ℹ]  deploying stack "eksctl-brain-tasks-cluster-nodegroup-brain-tasks-nodes"
2026-04-19 02:28:44 [ℹ]  waiting for CloudFormation stack "eksctl-brain-tasks-cluster-nodegroup-brain-tasks-nodes"
2026-04-19 02:29:16 [ℹ]  waiting for CloudFormation stack "eksctl-brain-tasks-cluster-nodegroup-brain-tasks-nodes"
2026-04-19 02:30:11 [ℹ]  waiting for CloudFormation stack "eksctl-brain-tasks-cluster-nodegroup-brain-tasks-nodes"
2026-04-19 02:31:07 [ℹ]  waiting for CloudFormation stack "eksctl-brain-tasks-cluster-nodegroup-brain-tasks-nodes"
2026-04-19 02:31:08 [ℹ]  waiting for the control plane to become ready
2026-04-19 02:31:10 [✔]  saved kubeconfig as "/home/abi/.kube/config"
2026-04-19 02:31:10 [ℹ]  no tasks
2026-04-19 02:31:10 [✔]  all EKS cluster resources for "brain-tasks-cluster" have been created
2026-04-19 02:31:12 [ℹ]  nodegroup "brain-tasks-nodes" has 2 node(s)
2026-04-19 02:31:12 [ℹ]  node "ip-192-168-46-119.ec2.internal" is ready
2026-04-19 02:31:12 [ℹ]  node "ip-192-168-6-115.ec2.internal" is ready
2026-04-19 02:31:12 [ℹ]  waiting for at least 1 node(s) to become ready in "brain-tasks-nodes"
2026-04-19 02:31:12 [ℹ]  nodegroup "brain-tasks-nodes" has 2 node(s)
2026-04-19 02:31:12 [ℹ]  node "ip-192-168-46-119.ec2.internal" is ready
2026-04-19 02:31:12 [ℹ]  node "ip-192-168-6-115.ec2.internal" is ready
2026-04-19 02:31:12 [✔]  created 1 managed nodegroup(s) in cluster "brain-tasks-cluster"
2026-04-19 02:31:15 [ℹ]  creating addon: metrics-server
2026-04-19 02:31:15 [ℹ]  successfully created addon: metrics-server
2026-04-19 02:31:19 [ℹ]  kubectl command should work with "/home/abi/.kube/config", try 'kubectl get nodes'
2026-04-19 02:31:19 [✔]  EKS cluster "brain-tasks-cluster" in "us-east-1" region is ready
vicky@ubuntu:~/Brain-Tasks-App$ kubectl get nodes
NAME                                           STATUS   ROLES    AGE     VERSION
ip-192-168-46-119.ec2.internal   Ready    <none>   3m35s   v1.34.6-eks-bbe087e
ip-192-168-6-115.ec2.internal    Ready    <none>   3m36s   v1.34.6-eks-bbe087e
```

```
kubectl get nodes
Updated context arn:aws:eks:us-east-1:839922332162:cluster/brain-tasks-cluster in /home/vicky/.kube/config
NAME                                           STATUS   ROLES    AGE     VERSION
ip-192-168-14-201.ec2.internal   Ready    <none>   7m17s   v1.34.4-eks-f69f56f
vicky@ubuntu:~/Brain-Tasks-App$ ls
deployment.yaml  dist  Dockerfile  README.md  service.yaml
vicky@ubuntu:~/Brain-Tasks-App$ cat deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: brain-tasks-deployment
spec:
  replicas: 1
  selector:
    matchLabels:
      app: brain-tasks
  template:
    metadata:
      labels:
        app: brain-tasks
    spec:
      containers:
      - name: brain-tasks
        image: 839922332162.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app:v1
        ports:
        - containerPort: 80
vicky@ubuntu:~/Brain-Tasks-App$ cat service.yaml
apiVersion: v1
kind: Service
metadata:
  name: brain-tasks-service
spec:
  selector:
    app: brain-tasks
  type: LoadBalancer
  ports:
  - protocol: TCP
    port: 80
    targetPort: 80
```

```
vicky@ubuntu:~/Brain-Tasks-App$ echo "=== Nodes ===" && \
kubectl get nodes && \
echo "=== Pods ===" && \
kubectl get pods && \
echo "=== Deployment ===" && \
kubectl get deployment && \
echo "=== Service ===" && \
kubectl get service brain-tasks-service
=== Nodes ===
NAME                                           STATUS   ROLES    AGE   VERSION
ip-192-168-14-201.ec2.internal   Ready    <none>   11m   v1.34.4-eks-f69f56f
=== Pods ===
NAME                                                READY   STATUS    RESTARTS   AGE
brain-tasks-deployment-f65d9bd97-chqkl   1/1     Running   0          2m35s
=== Deployment ===
NAME                                 READY   UP-TO-DATE   AVAILABLE   AGE
brain-tasks-deployment   1/1     1            1           2m39s
=== Service ===
NAME                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
TYPE           CLUSTER-IP       EXTERNAL-IP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
PORT(S)        AGE
brain-tasks-service   LoadBalancer   10.100.209.141   ad653e2fdfc8b4f3fb8f93f3401050ae-773780226.us-east-1.elb.amazonaws.com   80:32033/TCP   2m34s
vicky@ubuntu:~/Brain-Tasks-App$
```

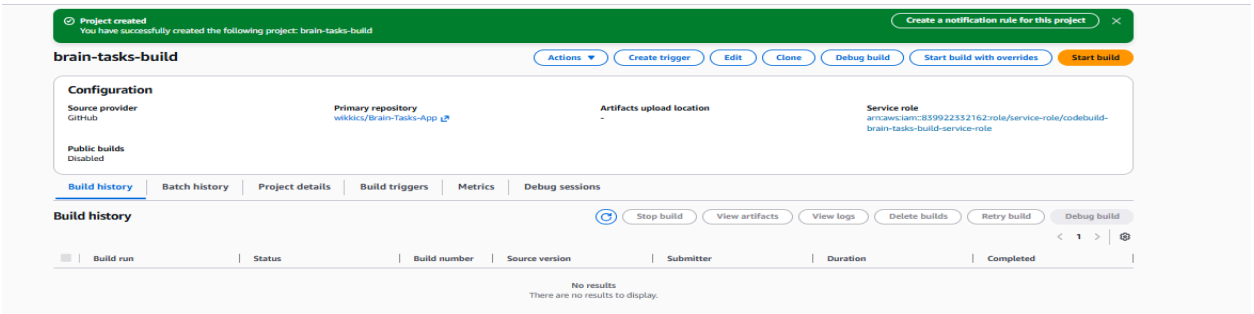


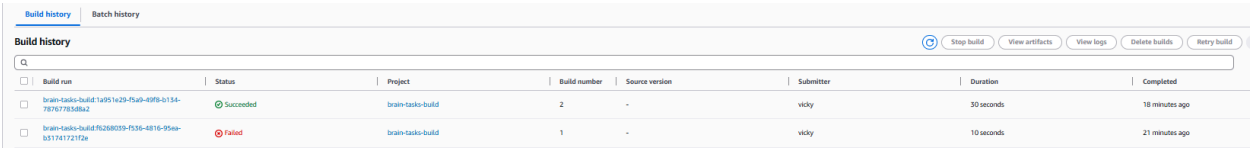

```
vicky@ubuntu:~/Brain-Tasks-App$ cd ~/Brain-Tasks-App

# Change remote to your forked repo
git remote set-url origin https://github.com/wikkics/Brain-Tasks-App.git

# Verify
git remote -v
origin  https://github.com/wikkics/Brain-Tasks-App.git (fetch)
origin  https://github.com/wikkics/Brain-Tasks-App.git (push)
vicky@ubuntu:~/Brain-Tasks-App$ git push origin main
Username for 'https://github.com': wikkics
Password for 'https://wikkics@github.com':
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 2 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 1.14 KiB | 290.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/wikkics/Brain-Tasks-App.git
   d611aeb..1b6afcd  main -> main
vicky@ubuntu:~/Brain-Tasks-App$
```

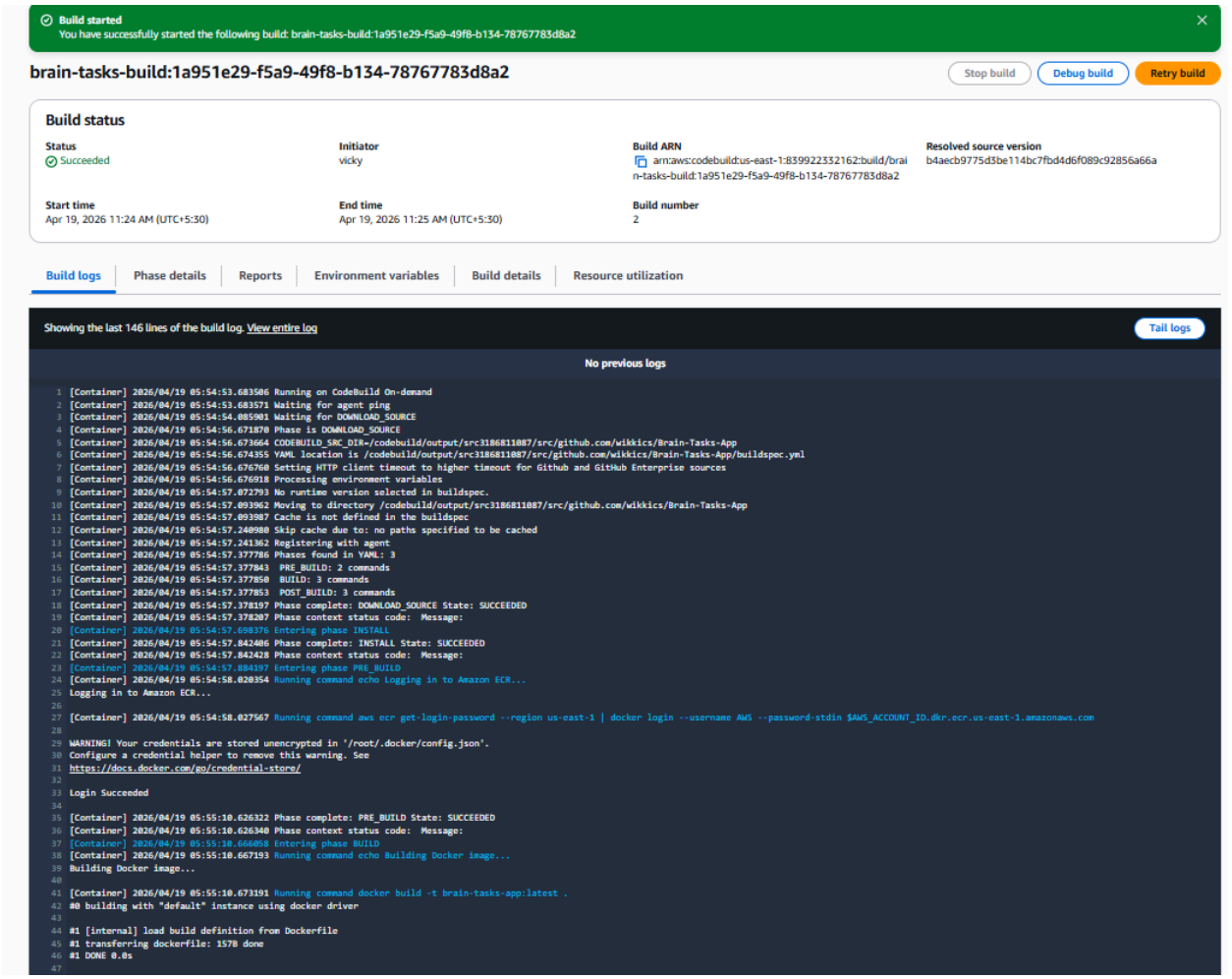


AWS CodeBuild (CI) :

Created a CodeBuild project that automatically builds the Docker image, pushes to ECR, and deploys to EKS. The buildspec.yml defines all build instructions.

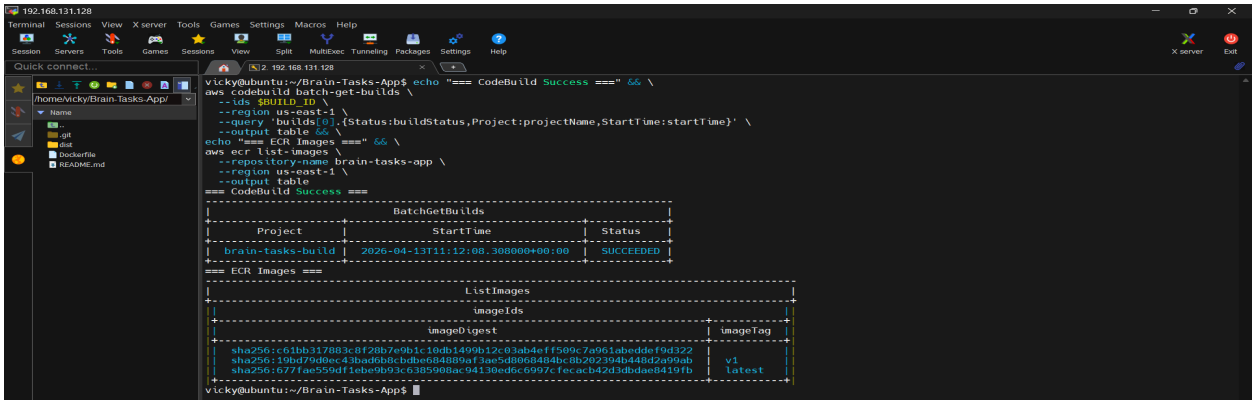






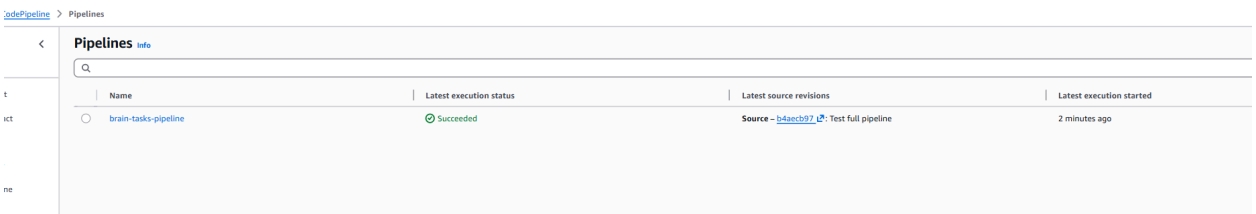
```
#1 [internal] load build definition from Dockerfile
#1 transferring dockerfile: 157B done
#1 DONE 0.0s

#2 [internal] load metadata for public.ecr.aws/nginx/nginx:alpine
#2 DONE 0.2s

#3 [internal] load .dockerignore
#3 transferring context: 2B done
#3 DONE 0.0s

#4 [internal] load build context
#4 transferring context: 317.98kB 0.0s done
#4 DONE 0.0s

#5 [1/2] FROM public.ecr.aws/nginx/nginx:alpine@sha256:72c9b856fee09b02f9e5ee29ab2b55ae9a3c3b59cf094baef1d05f5914d8cb17
#5 resolve public.ecr.aws/nginx/nginx:alpine@sha256:72c9b856fee09b02f9e5ee29ab2b55ae9a3c3b59cf094baef1d05f5914d8cb17 0.0s done
#5 extracting sha256:6a0ac1617861a677b045b7ff88545213ec31c0ff08763195a70a4a5adda577bb
#5 sha256:72c9b856fee09b02f9e5ee29ab2b55ae9a3c3b59cf094baef1d05f5914d8cb17 685B / 685B done
#5 sha256:a556523f89a2588511ab76c7b1fec6158c7ec5617e04365354a771d7b102020c 1.99kB / 1.99kB done
#5 sha256:6a0ac1617861a677b045b7ff88545213ec31c0ff08763195a70a4a5adda577bb 3.86MB / 3.86MB 0.1s done
#5 sha256:583599bb7d382e9e986a9ff6520498130 7bfa71670a29f4190eedae93c4858be 627B / 627B 0.1s done
#5 sha256:781ff50d2644f74714c0d5a4ffa3447bbfd4b293cfb1dc6a637722ae24c89241 404B / 404B 0.1s done
#5 sha256:453da7dbc73e7338c4e201ec3f3c4a5f7751adbf5a6f127792c1cd001a780721 0B / 1.21kB 0.1s
#5 sha256:4a8b0b2a5b1937755263ac7ac4ee26db2906c13a5d70f1fb1875ba8e370cdd8f 0B / 1.40kB 0.1s
#5 sha256:812d47f806db497c53f9b47e76bdab38bcf5d724c69d3986df5ee4336c210559 12.32kB / 12.32kB done
#5 sha256:82736a35d0e7f1309edc13d09115410b81542cf8b91afff80deeace02b6f2f26 1.87MB / 1.87MB 0.1s done
#5 sha256:aee4e54b3865ee4d98545b6b49f9e8ab3b9b6ed114a2eafedc7a22e984ea5a68 956B / 956B 0.1s done
#5 extracting sha256:6a0ac1617861a677b045b7ff88545213ec31c0ff08763195a70a4a5adda577bb 0.2s done
#5 sha256:453da7dbc73e7338c4e201ec3f3c4a5f7751adbf5a6f127792c1cd001a780721 1.21kB / 1.21kB 0.1s done
#5 sha256:4a8b0b2a5b1937755263ac7ac4ee26db2906c13a5d70f1fb1875ba8e370cdd8f 1.40kB / 1.40kB 0.1s done
#5 sha256:612c0c1df4c55a0bf145f84df03cb28de505e6a52fb8e49c9da7b143fc00ad2d 20.25MB / 20.25MB 0.3s done
#5 extracting sha256:82736a35d0e7f1309edc13d09115410b81542cf8b91afff80deeace02b6f2f26 0.1s done
#5 extracting sha256:583599bb7d382e9e986a9ff6520498130 7bfa71670a29f4190eedae93c4858be
#5 extracting sha256:583599bb7d382e9e986a9ff6520498130 7bfa71670a29f4190eedae93c4858be done
#5 extracting sha256:aee4e54b3865ee4d98545b6b49f9e8ab3b9b6ed114a2eafedc7a22e984ea5a68 done
#5 extracting sha256:781ff50d2644f74714c0d5a4ffa3447bbfd4b293cfb1dc6a637722ae24c89241 done
#5 extracting sha256:453da7dbc73e7338c4e201ec3f3c4a5f7751adbf5a6f127792c1cd001a780721
#5 extracting sha256:453da7dbc73e7338c4e201ec3f3c4a5f7751adbf5a6f127792c1cd001a780721 done
#5 extracting sha256:4a8b0b2a5b1937755263ac7ac4ee26db2906c13a5d70f1fb1875ba8e370cdd8f done
#5 extracting sha256:612c0c1df4c55a0bf145f84df03cb28de505e6a52fb8e49c9da7b143fc00ad2d 0.1s
#5 extracting sha256:612c0c1df4c55a0bf145f84df03cb28de505e6a52fb8e49c9da7b143fc00ad2d 0.4s done
#5 DONE 1.3s

#6 [2/2] COPY dist/ /usr/share/nginx/html
#6 DONE 0.2s

#7 exporting to image
#7 exporting layers 0.0s done
#7 writing image sha256:ecc903dfbec7fe067c76b12e5761c435edd9ac6d328bbad63da7fd9a69147c5c done
#7 naming to docker.io/library/brain-tasks-app:latest done
#7 DONE 0.0s

[Container] 2026/04/19 05:55:12.808632 Running command docker tag brain-tasks-app:latest $AWS_ACCOUNT_ID.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app:latest

[Container] 2026/04/19 05:55:12.834450 Phase complete: BUILD State: SUCCEEDED
[Container] 2026/04/19 05:55:12.834473 Phase context status code:  Message:
[Container] 2026/04/19 05:55:12.875522 Entering phase POST_BUILD
[Container] 2026/04/19 05:55:12.877039 Running command echo Pushing image to ECR...
Pushing image to ECR...

[Container] 2026/04/19 05:55:12.882924 Running command docker push $AWS_ACCOUNT_ID.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app:latest
The push refers to repository [839922332162.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app]
842e895e4c42: Preparing
36c003a300d1: Preparing
2b12ccfb247a: Preparing
646d544dc619: Preparing
f7b74afe8798: Preparing
00a43f630638: Preparing
f4eddaa4c303: Preparing
e6082cdc2516: Preparing
29df493baa13: Preparing
00a43f630638: Waiting
f4eddaa4c303: Waiting
e6082cdc2516: Waiting
29df493baa13: Waiting
36c003a300d1: Layer already exists
```

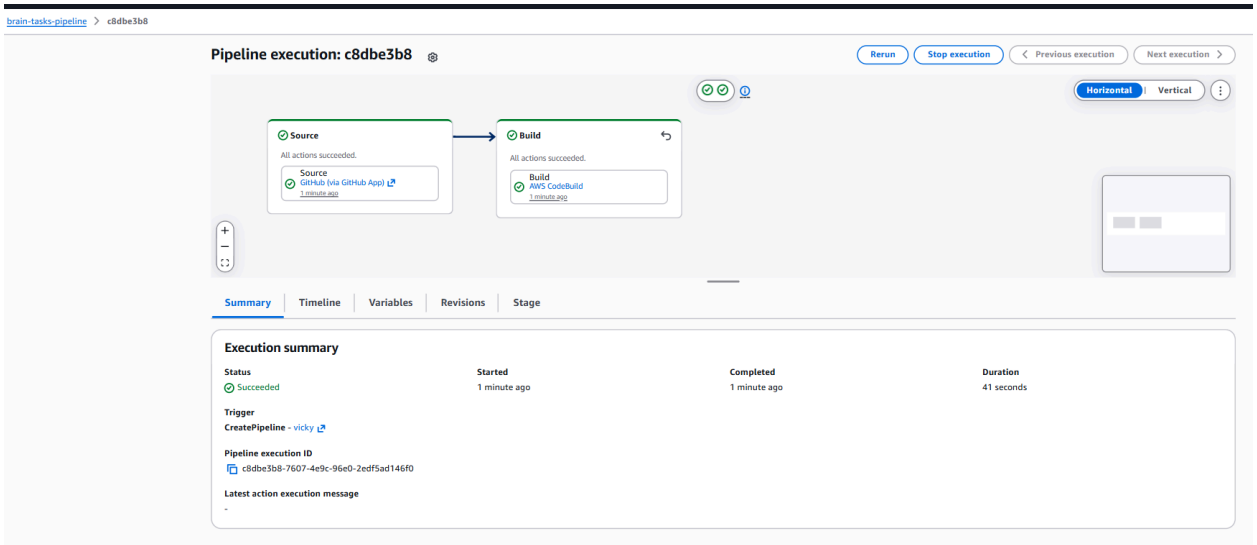
```
[Container] 2026/04/19 05:55:12.882924 Running command docker push $AWS_ACCOUNT_ID.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app:latest
The push refers to repository [839922332162.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app]
842e895e4c42: Preparing
36c003a300d1: Preparing
2b12ccfb247a: Preparing
646d544dc619: Preparing
f7b74afe8798: Preparing
00a43f630638: Preparing
f4eddaa4c303: Preparing
e6082cdc2516: Preparing
29df493baa13: Preparing
00a43f630638: Waiting
f4eddaa4c303: Waiting
e6082cdc2516: Waiting
29df493baa13: Waiting
36c003a300d1: Layer already exists
646d544dc619: Layer already exists
f7b74afe8798: Layer already exists
2b12ccfb247a: Layer already exists
00a43f630638: Layer already exists
f4eddaa4c303: Layer already exists
e6082cdc2516: Layer already exists
29df493baa13: Layer already exists
842e895e4c42: Pushed
latest: digest: sha256:9a380335643fa55ab70f5264c1ae8d1c28174ffc460a83e11b3aa6e168a9de57 size: 2199

[Container] 2026/04/19 05:55:14.257498 Running command echo Build complete!
Build complete!

[Container] 2026/04/19 05:55:14.263598 Phase complete: POST_BUILD State: SUCCEEDED
[Container] 2026/04/19 05:55:14.263616 Phase context status code:  Message:
[Container] 2026/04/19 05:55:14.317034 Set report auto-discover timeout to 5 seconds
[Container] 2026/04/19 05:55:14.317077 Expanding base directory path:  .
[Container] 2026/04/19 05:55:14.318677 Assembling file list
[Container] 2026/04/19 05:55:14.318690 Expanding .
[Container] 2026/04/19 05:55:14.320380 Expanding file paths for base directory .
[Container] 2026/04/19 05:55:14.320389 Assembling file list
[Container] 2026/04/19 05:55:14.320396 Expanding **/*
[Container] 2026/04/19 05:55:14.322467 No matching auto-discover report paths found
[Container] 2026/04/19 05:55:14.322562 Report auto-discover file discovery took 0.005527 seconds
[Container] 2026/04/19 05:55:14.322576 Phase complete: UPLOAD_ARTIFACTS State: SUCCEEDED
[Container] 2026/04/19 05:55:14.322582 Phase context status code:  Message:
```



AWS CodePipeline (CD) :

Created a two-stage pipeline connecting GitHub to CodeBuild. Every push to the main branch automatically triggers a full build and deployment.





Pipeline execution: c8dbe3b8

← Overvie

▶ Pipeline execution details

Logs | Summary | Input | Output

Showing the last 149 lines of the build log. View entire log

Tail logs

No previous logs

```
1  [Container] 2026/04/19 06:40:22.817872 Running on CodeBuild On-demand
2  [Container] 2026/04/19 06:40:22.817984 Waiting for agent ping
3  [Container] 2026/04/19 06:40:23.219858 Waiting for DOWNLOAD_SOURCE
4  [Container] 2026/04/19 06:40:24.759238 Phase is DOWNLOAD_SOURCE
5  [Container] 2026/04/19 06:40:24.760453 CODEBUILD_SRC_DIR=/codebuild/output/src815325292/src
6  [Container] 2026/04/19 06:40:24.760951 YAML location is /codebuild/output/src815325292/src/buildspec.yml
7  [Container] 2026/04/19 06:40:24.763327 Setting HTTP client timeout to higher timeout for S3 source
8  [Container] 2026/04/19 06:40:24.763417 Processing environment variables
9  [Container] 2026/04/19 06:40:25.184613 No runtime version selected in buildspec.
10 [Container] 2026/04/19 06:40:25.202801 Moving to directory /codebuild/output/src815325292/src
11 [Container] 2026/04/19 06:40:25.202823 Cache is not defined in the buildspec
12 [Container] 2026/04/19 06:40:25.367646 Skip cache due to: no paths specified to be cached
13 [Container] 2026/04/19 06:40:25.367927 Registering with agent
14 [Container] 2026/04/19 06:40:25.504188 Phases found in YAML: 3
15 [Container] 2026/04/19 06:40:25.504123  PRE_BUILD: 2 commands
16 [Container] 2026/04/19 06:40:25.504128  BUILD: 3 commands
17 [Container] 2026/04/19 06:40:25.504132  POST_BUILD: 3 commands
18 [Container] 2026/04/19 06:40:25.504560 Phase complete: DOWNLOAD_SOURCE State: SUCCEEDED
19 [Container] 2026/04/19 06:40:25.504573 Phase context status code:  Message:
20 [Container] 2026/04/19 06:40:25.810523 Entering phase INSTALL
21 [Container] 2026/04/19 06:40:25.949432 Phase complete: INSTALL State: SUCCEEDED
22 [Container] 2026/04/19 06:40:25.949448 Phase context status code:  Message:
23 [Container] 2026/04/19 06:40:25.992164 Entering phase PRE_BUILD
24 [Container] 2026/04/19 06:40:26.142871 Running command echo Logging in to Amazon ECR...
25 Logging in to Amazon ECR...
26
27 [Container] 2026/04/19 06:40:26.149311 Running command aws ecr get-login-password --region us-east-1 | docker login --username AWS --password-stdin $AWS_ACCOUNT_ID.dkr.ecr.us-east-1.amazonaws.com
28
29 WARNING! Your credentials are stored unencrypted in '/root/.docker/config.json'.
30 Configure a credential helper to remove this warning. See
31 https://docs.docker.com/go/credential-store/
32
33 Login Succeeded
34
35 [Container] 2026/04/19 06:40:38.302140 Phase complete: PRE_BUILD State: SUCCEEDED
36 [Container] 2026/04/19 06:40:38.302159 Phase context status code:  Message:
37 [Container] 2026/04/19 06:40:38.337176 Entering phase BUILD
38 [Container] 2026/04/19 06:40:38.338526 Running command echo Building Docker image...
39 Building Docker image...
40
41 [Container] 2026/04/19 06:40:38.344960 Running command docker build -t brain-tasks-app:latest .
42 #0 building with "default" instance using docker driver
43
44 #1 [internal] load build definition from Dockerfile
45 #1 transferring dockerfile: 157B done
46 #1 DONE 0.0s
47
48 #2 [internal] load metadata for public.ecr.aws/nginx/nginx:alpine
49 #2 DONE 0.2s
50
51 #3 [internal] load .dockerignore
52 #3 transferring context: 2B done
53 #3 DONE 0.0s
54
55 #4 [internal] load build context
56 #4 transferring context: 317.98kB 0.0s done
57 #4 DONE 0.0s
58
```

Pipeline execution: c8dbe3b8

← Over

▶ **Pipeline execution details**

Logs | Summary | Input | Output

```
#3 [internal] load .dockerignore
#3 transferring context: 2B done
#3 DONE 0.0s

#4 [internal] load build context
#4 transferring context: 317.98kB 0.0s done
#4 DONE 0.0s

#5 [1/2] FROM public.ecr.aws/nginx/nginx:alpine@sha256:72c9b856fee09b02f9e5ee29ab2b55ae9a3c3b59cf094baef1d05f5914d8cb17
#5 resolve public.ecr.aws/nginx/nginx:alpine@sha256:72c9b856fee09b02f9e5ee29ab2b55ae9a3c3b59cf094baef1d05f5914d8cb17 0.0s done
#5 sha256:72c9b856fee09b02f9e5ee29ab2b55ae9a3c3b59cf094baef1d05f5914d8cb17 685B / 685B done
#5 sha256:82736a35d0e7f1309edc13d09115410b81542cf8b91aff80deeace02b6f2f26 1.87MB / 1.87MB 0.1s done
#5 sha256:583599bb7d382e9e986a9ff65204981307bfa71670a29f4190eedae93c4858be 627B / 627B 0.1s done
#5 sha256:781ff50d2644f74714c0d5a4ffa3447bbfd4b293cfb1dc6a637722ae24c89241 0B / 404B 0.1s
#5 sha256:453da7dbc73e7338c4e201ec3f3c4a5f7751adbf5a6f127792c1cd001a780721 0B / 1.21kB 0.1s
#5 sha256:4a8b0b2a5b1937755263ac7ac4ee26db2906c13a5d70f1fb1875ba8e370cdd8f 0B / 1.40kB 0.1s
#5 sha256:a556523f89a2588511ab76c7b1fec6158c7ec5617e04365354a771d7b102020c 1.99kB / 1.99kB done
#5 sha256:812d47f806db497c53f9b47e76bdab38bcf5d724c69d3986df5ee4336c210559 12.32kB / 12.32kB done
#5 sha256:6a0ac1617861a677b045b7ff88545213ec31c0ff08763195a70a4a5adda577bb 3.86MB / 3.86MB 0.1s done
#5 sha256:aee4e54b3865ee4d98545b6b49f9e8ab3b9b6ed114a2eafedc7a22e984ea5a68 956B / 956B 0.1s done
#5 sha256:781ff50d2644f74714c0d5a4ffa3447bbfd4b293cfb1dc6a637722ae24c89241 404B / 404B 0.1s done
#5 sha256:453da7dbc73e7338c4e201ec3f3c4a5f7751adbf5a6f127792c1cd001a780721 1.21kB / 1.21kB 0.1s done
#5 sha256:4a8b0b2a5b1937755263ac7ac4ee26db2906c13a5d70f1fb1875ba8e370cdd8f 1.40kB / 1.40kB 0.1s done
#5 extracting sha256:6a0ac1617861a677b045b7ff88545213ec31c0ff08763195a70a4a5adda577bb
#5 sha256:612c0c1df4c55a0bf145f84df03cb28de505e6a52fb8e49c9da7b143fc00ad2d 10.49MB / 20.25MB 0.2s
#5 extracting sha256:6a0ac1617861a677b045b7ff88545213ec31c0ff08763195a70a4a5adda577bb 0.2s done
#5 sha256:612c0c1df4c55a0bf145f84df03cb28de505e6a52fb8e49c9da7b143fc00ad2d 20.25MB / 20.25MB 0.3s done
#5 extracting sha256:82736a35d0e7f1309edc13d09115410b81542cf8b91aff80deeace02b6f2f26
#5 extracting sha256:82736a35d0e7f1309edc13d09115410b81542cf8b91aff80deeace02b6f2f26 0.1s done
#5 extracting sha256:583599bb7d382e9e986a9ff65204981307bfa71670a29f4190eedae93c4858be
#5 extracting sha256:583599bb7d382e9e986a9ff65204981307bfa71670a29f4190eedae93c4858be done
#5 extracting sha256:aee4e54b3865ee4d98545b6b49f9e8ab3b9b6ed114a2eafedc7a22e984ea5a68 done
#5 extracting sha256:781ff50d2644f74714c0d5a4ffa3447bbfd4b293cfb1dc6a637722ae24c89241 done
#5 extracting sha256:453da7dbc73e7338c4e201ec3f3c4a5f7751adbf5a6f127792c1cd001a780721
#5 extracting sha256:453da7dbc73e7338c4e201ec3f3c4a5f7751adbf5a6f127792c1cd001a780721 done
#5 extracting sha256:4a8b0b2a5b1937755263ac7ac4ee26db2906c13a5d70f1fb1875ba8e370cdd8f done
#5 extracting sha256:612c0c1df4c55a0bf145f84df03cb28de505e6a52fb8e49c9da7b143fc00ad2d
#5 extracting sha256:612c0c1df4c55a0bf145f84df03cb28de505e6a52fb8e49c9da7b143fc00ad2d 0.5s done
#5 DONE 1.4s

#6 [2/2] COPY dist/ /usr/share/nginx/html
#6 DONE 0.3s

#7 exporting to image
#7 exporting layers 0.0s done
#7 writing image sha256:b321f7abd41d4b2ec839cae4e8a533abdb18305dd9785c3ce0c8bb3658b7e613 done
#7 naming to docker.io/library/brain-tasks-app:latest done
#7 DONE 0.0s

[Container] 2026/04/19 06:40:40.644509 Running command docker tag brain-tasks-app:latest $AWS_ACCOUNT_ID.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app:latest

[Container] 2026/04/19 06:40:40.666639 Phase complete: BUILD State: SUCCEEDED
[Container] 2026/04/19 06:40:40.666656 Phase context status code:  Message:
[Container] 2026/04/19 06:40:40.721925 Entering phase POST_BUILD
[Container] 2026/04/19 06:40:40.723082 Running command echo Pushing image to ECR...
Pushing image to ECR...

[Container] 2026/04/19 06:40:40.729397 Running command docker push $AWS_ACCOUNT_ID.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app:latest
The push refers to repository [839922332162.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app]
266c34a6f31a: Preparing
36c003a300d1: Preparing
2b12ccfb247a: Preparing
646d544dc619: Preparing
```

```
[Container] 2026/04/19 06:40:40.729397 Running command docker push $AWS_ACCOUNT_ID.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app:latest
The push refers to repository [839922332162.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app]
266c34a6f31a: Preparing
36c003a300d1: Preparing
2b12ccfb247a: Preparing
646d544dc619: Preparing
f7b74afe8798: Preparing
00a43f630638: Preparing
f4eddaa4c303: Preparing
e6082cdc2516: Preparing
29df493baa13: Preparing
00a43f630638: Waiting
f4eddaa4c303: Waiting
e6082cdc2516: Waiting
29df493baa13: Waiting
646d544dc619: Layer already exists
36c003a300d1: Layer already exists
f7b74afe8798: Layer already exists
2b12ccfb247a: Layer already exists
00a43f630638: Layer already exists
29df493baa13: Layer already exists
e6082cdc2516: Layer already exists
f4eddaa4c303: Layer already exists
266c34a6f31a: Pushed
latest: digest: sha256:fe76a894d6e55296d963eb02179e328adff79cfb55a5243da33f68d91354ed2b size: 2199

[Container] 2026/04/19 06:40:42.155607 Running command echo Build complete!
Build complete!

[Container] 2026/04/19 06:40:42.162139 Phase complete: POST_BUILD State: SUCCEEDED
[Container] 2026/04/19 06:40:42.162155 Phase context status code:  Message:
[Container] 2026/04/19 06:40:42.211356 Set report auto-discover timeout to 5 seconds
[Container] 2026/04/19 06:40:42.211398 Expanding base directory path:  .
[Container] 2026/04/19 06:40:42.213443 Assembling file list
[Container] 2026/04/19 06:40:42.213462 Expanding .
[Container] 2026/04/19 06:40:42.215442 Expanding file paths for base directory .
[Container] 2026/04/19 06:40:42.215453 Assembling file list
[Container] 2026/04/19 06:40:42.215457 Expanding **/*
[Container] 2026/04/19 06:40:42.217578 No matching auto-discover report paths found
[Container] 2026/04/19 06:40:42.217594 Report auto-discover file discovery took 0.006239 seconds
[Container] 2026/04/19 06:40:42.217604 Phase complete: UPLOAD_ARTIFACTS State: SUCCEEDED
[Container] 2026/04/19 06:40:42.217610 Phase context status code:  Message:
```

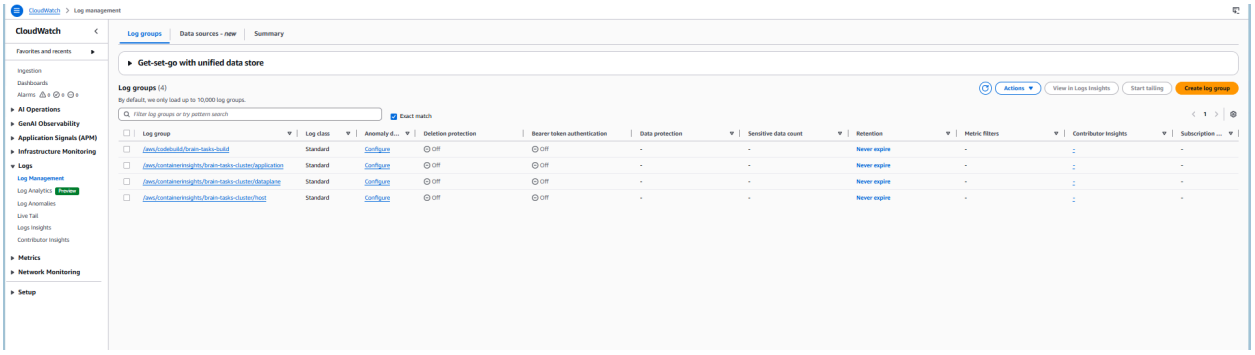
```
192.168.131.128
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help X server Exit
Quick connect...
/home/vicky/Brain-Tasks-App/
Name
..
.git
dist
Dockerfile
README.md
Remote monitoring
2. 192.168.131.128
  --output table
=== App is Live ===
NAME                                       STATUS   ROLES    AGE   VERSION
ip-192-168-14-201.ec2.internal             Ready    <none>   82m   v1.34.4-eks-f69f56f
NAME                                       READY   STATUS    RESTARTS   AGE
brain-tasks-deployment-f65d9bd97-chqkl     1/1     Running   0          72m
NAME                 TYPE           CLUSTER-IP       EXTERNAL-IP                                                              PORT(S)        AGE
brain-tasks-service  LoadBalancer   10.100.209.141   ad653e2fdfc8b4f3fb8f93f3401050ae-773780226.us-east-1.elb.amazonaws.com   80:32033/TCP   72m

=== Pipeline ===
------------------------
|   GetPipelineState   |
+---------+------------+
|  Stage  |   Status   |
+---------+------------+
|  Source |  Succeeded |
|  Build  |  Succeeded |
+---------+------------+
=== ECR Image s ===
-----------------------------------------------------------------------------------------
|                                        ListImages                                       |
+-----------------------------------------------------------------------------------------+
||                                          imageIds                                       ||
|+-----------------------------------------------------------------------+-----------+|
||                                imageDigest                            | imageTag  ||
|+-----------------------------------------------------------------------+-----------+|
||  sha256:c61bb317883c8f28b7e9b1c10db1499b12c03ab4eff509c7a961abeddef9d322  |           ||
||  sha256:677fae559df1ebe9b93c6385908ac94130ed6c6997cfecacb42d3dbdae8419fb  |           ||
||  sha256:47b21e3408d230acfee0345744775c2ee18c9eee68ef900316c594daab2c47a2  |           ||
||  sha256:01cafcccf1d0a74975db5d4e4bb93849129d91327ac4dd0f13e6438291166232  |           ||
||  sha256:e28e157e6421653e75425030813a262f8bbb1650cd96ed4c04ced3957c226a04  |  latest   ||
||  sha256:19bd79d0ec43bad6b8cbdbe684889af3ae5d8068484bc8b202394b448d2a99ab  |  v1       ||
|+-----------------------------------------------------------------------+-----------+|
=== CloudWatch Logs ===
-------------------------------------
|          DescribeLogGroups        |
+-----------------------------------+
|  /aws/codebuild/brain-tasks-build |
+-----------------------------------+
vicky@ubuntu:~/Brain-Tasks-App$
```

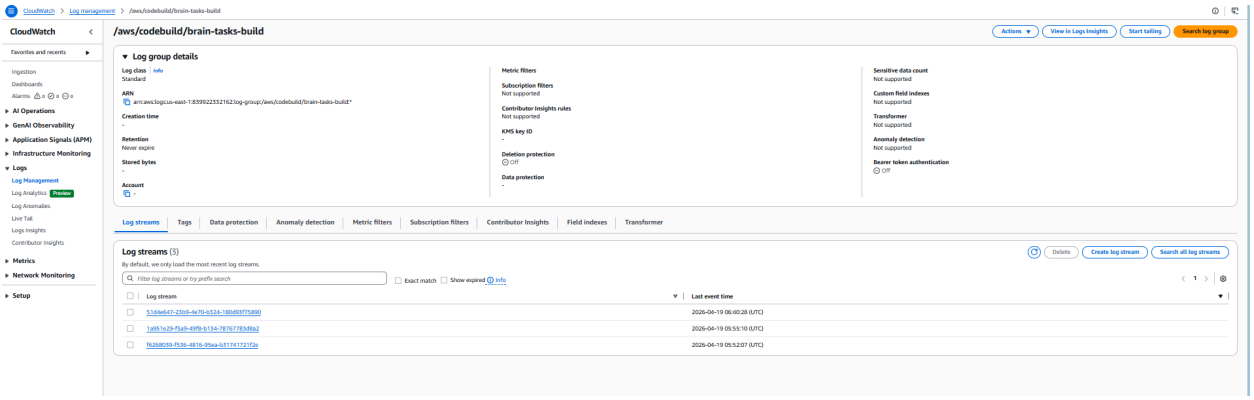
Monitoring:Cloud Watch

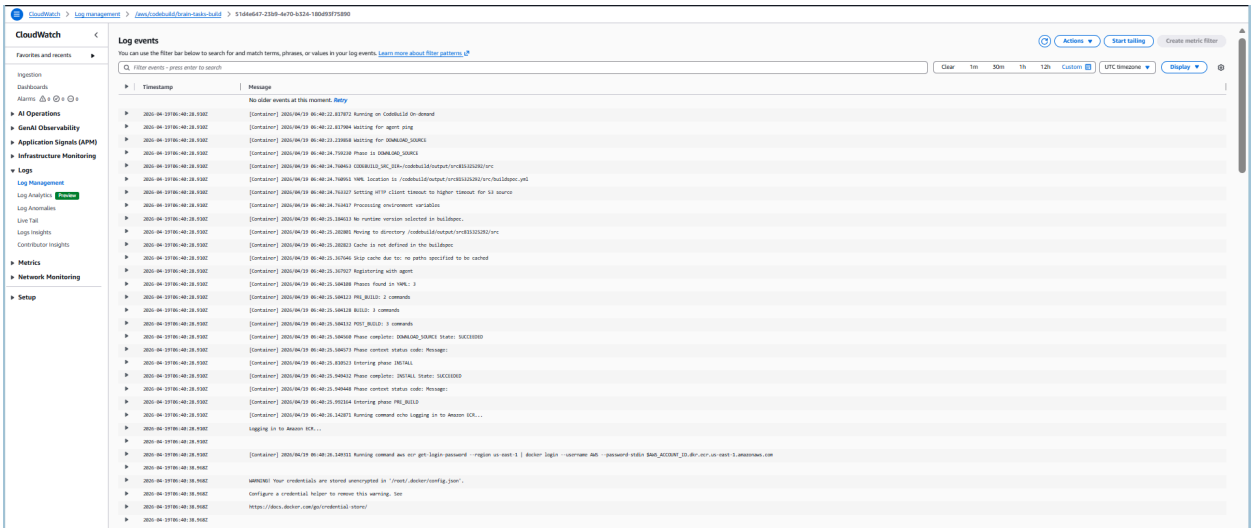
I have deployed Fluent Bit as a DaemonSet on EKS to stream application and node logs to AWS CloudWatch, alongside automatic CodeBuild logs resulting in 4 log groups tracking build, application, dataplane, and host activity

```
vicky@ubuntu:~/Brain-Tasks-App$ kubectl apply -f https://raw.githubusercontent.com/aws-samples/amazon-cloudwatch-container-insights/latest/k8s-deployment-manifest-templates/deployment-mode/daemonset/container-insights-monitoring/cloudwatch-namespace.yaml

kubectl create configmap fluent-bit-cluster-info \
  --from-literal=cluster.name=brain-tasks-cluster \
  --from-literal=http.server=On \
  --from-literal=http.port=2020 \
  --from-literal=read.head=Off \
  --from-literal=read.tail=On \
  --from-literal=logs.region=us-east-1 \
  -n amazon-cloudwatch

kubectl apply -f https://raw.githubusercontent.com/aws-samples/amazon-cloudwatch-container-insights/latest/k8s-deployment-manifest-templates/deployment-mode/daemonset/container-insights-monitoring/fluent-bit/fluent-bit.yaml
namespace/amazon-cloudwatch created
configmap/fluent-bit-cluster-info created
serviceaccount/fluent-bit created
clusterrole.rbac.authorization.k8s.io/fluent-bit-role created
clusterrolebinding.rbac.authorization.k8s.io/fluent-bit-role-binding created
configmap/fluent-bit-config created
daemonset.apps/fluent-bit created
vicky@ubuntu:~/Brain-Tasks-App$
```

```
kubectl apply -f https://raw.githubusercontent.com/aws-samples/amazon-cloudwatch-container-insights/latest/k8s-deployment-manifest-templates/deployment-mode/daemonset/container-insights-monitoring/fluent-bit/fluent-bit.yaml
namespace/amazon-cloudwatch created
configmap/fluent-bit-cluster-info created
serviceaccount/fluent-bit created
clusterrole.rbac.authorization.k8s.io/fluent-bit-role created
clusterrolebinding.rbac.authorization.k8s.io/fluent-bit-role-binding created
configmap/fluent-bit-config created
daemonset.apps/fluent-bit created
vicky@ubuntu:~/Brain-Tasks-App$ aws iam attach-role-policy \
  --role-name eksctl-brain-tasks-cluster-nodegro-NodeInstanceRole-C0yQVLZzf9Fl \
  --policy-arn arn:aws:iam::aws:policy/CloudWatchAgentServerPolicy
vicky@ubuntu:~/Brain-Tasks-App$ kubectl get pods -n amazon-cloudwatch
NAME              READY   STATUS    RESTARTS   AGE
fluent-bit-6sl8m  0/1     Pending   0          4m32s
fluent-bit-cx94t  1/1     Running   0          4m32s
fluent-bit-l4spw  0/1     Pending   0          4m32s
fluent-bit-wsqmg  1/1     Running   0          4m32s
vicky@ubuntu:~/Brain-Tasks-App$
```

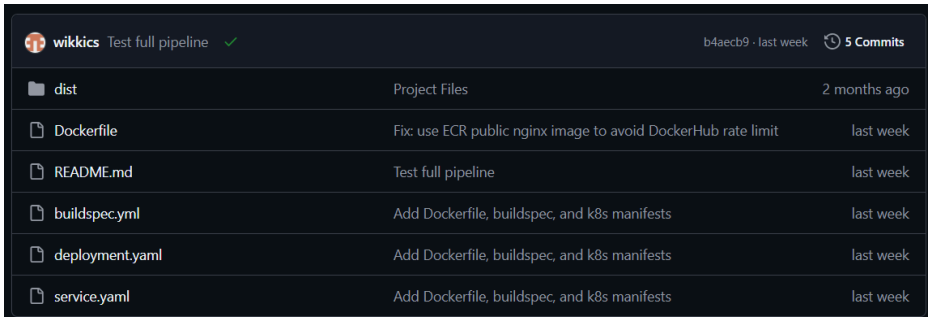
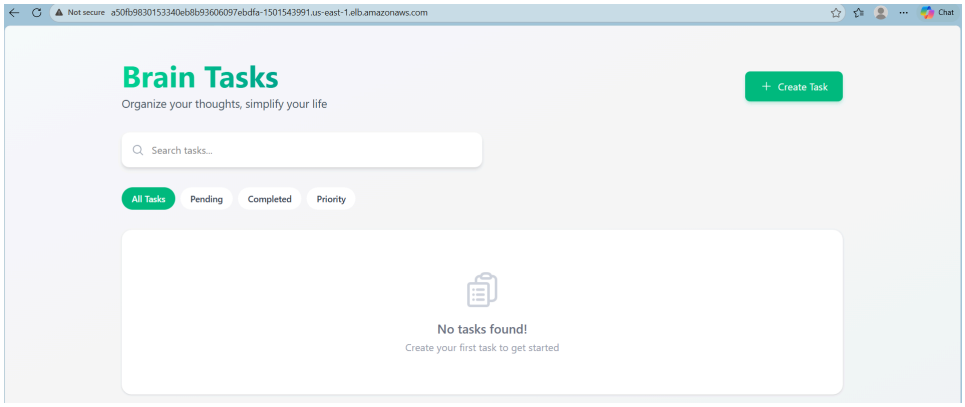







```
vicky@ubuntu:~/Brain-Tasks-App$ kubectl get pods -n amazon-cloudwatch
NAME                    READY   STATUS    RESTARTS   AGE
fluent-bit-6sl8m        0/1     Pending   0          4m32s
fluent-bit-cx94t        1/1     Running   0          4m32s
fluent-bit-l4spw        0/1     Pending   0          4m32s
fluent-bit-wsqmg        1/1     Running   0          4m32s
vicky@ubuntu:~/Brain-Tasks-App$ kubectl get svc -n default
NAME                 TYPE           CLUSTER-IP       EXTERNAL-IP                                                              PORT(S)        AGE
brain-tasks-service  LoadBalancer   10.100.165.205   a50fb9830153340eb8b93606097ebdfa-1501543991.us-east-1.elb.amazonaws.com  80:30370/TCP   4h24m
kubernetes           ClusterIP      10.100.0.1       <none>                                                                   443/TCP        4h37m
vicky@ubuntu:~/Brain-Tasks-App$
```

Repository	https://github.com/wikkics/Brain-Tasks-App



Application Type	React Web Application
Deployment Platform	Kubernetes (AWS EKS / EC2-based cluster assumed)
Containerization	Docker
CI/CD Tool	AWS CodeBuild
Web Server	Nginx
Live URL	http://a50fb9830153340eb8b93606097ebdfa-1501543991.us-east-1.elb.amazonaws.com